Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/13/2005
PI Name:	Drysdale, Alan E Ph.D.		
Project Title:	Systems Impacts of Waste Processing		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32815	**Congressional District:**	24
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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	Solid waste processing has potential and unavoidable impacts on all life support sub-systems. Either waste is produced that must be disposed of, waste processing has impacts on other sub-systems that must be considered, or waste is usable to reduce the cost of supporting the mission. The crew produce both trash and metabolic waste, food crops produce large quantities of inedible biomass, all systems produce used spares and consumables, and processing of the various kinds of waste typically produce particulate and trace gas contaminants, as well as producing or using quantities of water. Yet we are not aware of a quantitative study of the system impacts of solid waste so that objective decisions can be made about the best way to deal with the waste on a particular mission. We propose to identify life-support-related solid waste streams, options for processing the waste, costs and benefits, and recommend particular approaches for particular near-term missions.		

Task Description:	Our approach will continue to be built around equivalent system mass (ESM), as has previous life-support work we have done over the past decade. We will define expected waste streams for long duration missions based on historical and current data. This will involve working closely with KSC STS and ISS personnel, as there is no waste monitoring program currently in place for these programs. We will investigate data on earlier programs also. Based on these data we will develop waste stream models for the missions defined in the SMAP Reference Missions Document. This will involve extrapolation to longer missions and for different environments. We will develop waste processing scenarios ranging from simple dumping to complete destruction of waste. We will estimate ESM impacts of the various technologies, and assess mission ESM impacts based on generally agreed assumptions about infrastructure. The work will be coordinated through SMAP, and with the rest of the life support community. Attention will be paid to planetary protection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	May identify technologies significant for terrestrial use.
Task Progress:	Waste processing was compared for terrestrial and space applications. Typical quantities of terrestrial domestic municipal waste were about 0.6 kg/person per day, with sewage being collected separately and gaseous emissions typically not considered at all. The standard waste production models for space missions that were developed previously were expanded to include new exploration missions and non-metabolic wastes. Typical rates varied from 1.8 to 3.5 kg/md. Potential life support contingencies and probable consequences for the waste system were identified. In general, contingencies would increase waste loads, sometimes by orders of magnitude. New types of wastes could be created that would be challenging to deal with. For the Mars transit, surface, and base waste models, each of the types of waste was classified by proportion of 5 categories: ash, equipment, organic materials (biodegradable), plastic (non-biodegradable organics), and water content. This data was used to determine sizing of equipment in calculating ESM. Missions were tracked, including the new exploration missions, and appropriate equivalencies were determined for trade studies. Technology drivers were identified and their impacts on waste processing considered. Equivalencies are anticipated to fall due to development of infrastructure technologies over the next several decades as exploration missions take us further from Earth. Waste processing technologies were identified for Skylab, STS, and ISS missions, and from recent ALS projects. Waste technology interfaces were identified. Specific performances were calculated for the different technologies. Results varied from 1.7 kg processed per kg ESM for lyophilization on a transit mission to 7,000 for compacting on a Mars base. However, interface impacts could not be considered on a technology basis. Pre- and post- processing was in some cases as large an impact as the primary processing technology. Waste approaches were optimized using ESM as the primary metric. Interfaces were a driver in trade studies especially where oxidative processes were considered. Water balance was a major factor in determining the optimum life support system. Water balance varied depending on the technologies used, from a loss of about 5 kg per crewperson per day for an open loop system, to a gain of about 0.7 kg with the maximum water recovery from waste. EVA could result in a loss of about 0.4 kg/md with current EVA technologies and a Mars surface EVA model. Bioregeneration reduces the amount of waste and supply needed, but can have an unfavorable effect on water balance as less water is imported in food. Thus, any losses such as for EVA become more important. Some improvements in ALS reference missions were identified. Significant coordination was done. This included supporting the Habitation Conference, ICES, and COSPAR (editorial work from a previous year), small businesses (e.g. Orbitec), SIMA, the NASA solid waste group, and universities. University support included the PI serving on the external advisory board for the life support NSCORT led by Purdue and the ES CSTC at UF, Gainesville. Work with SIMA included assessing the application of ESM to trade studies, and the effects of uncertainties and gaps in the data.
Bibliography Type:	Description: (Last Updated: 04/14/2021)
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